

Anushka Dongaonkar

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Education

Boston University, Boston

Sep 2024 - Dec 2025

MS in Bioinformatics

GPA - 3.91

Vellore Institute of Technology, India

Sep 2020 - May 2024

Bachelors in Biotechnology

GPA - 8.75

Experience

Zhang Lab, Boston University

Oct 2024 - Current

Researcher

- Analyzing proteomics and metabolomics data from Alzheimer's Disease Neuroimaging Initiative (ADNI).
- Applying supervised machine learning to identify biomarkers for mild cognitive impairment and AD progression.

Skills - Python, R, Machine Learning, Data Integration, Alzheimer's Disease, Data Visualization

Boston University

Jan 2025 - Mar 2025

Teaching Assistant

- Designed and graded assignments and projects for the course *Unsupervised and Unstructured Machine Learning*

Skills: Unsupervised Machine Learning, Data Preparation, Mentorship

CSIR-Institute of Microbial Technology, India

Jan 2024 - June 2024

Research Trainee

- Developed a network-based drug repurposing pipeline that predicted 373 candidate compounds for combating antibiotic-resistant *Pseudomonas aeruginosa*.
- Handled Drug-Target Integration network data processing, annotation, and visualization.

Skills - Python, Linux, Big Data, Data cleaning, Data visualization

NeuroGen Brain and Spinal Research Institute, India

May 2023 - June 2023

Clinical Research Intern

- Compiled and analyzed clinical data to assess the impact of anti-aging therapy on biological age for 30 study participants.

Skills - Clinical Data Analysis, Scientific Writing, MS Office

Protein Biophysics Lab, IIT-Madras, India

May 2023 - Sep 2023

Research Intern

- Analyzed the relationship between protein size and length using MATLAB to enhance understanding of structural properties in proteins from the PDB database

Skills - MATLAB, Protein Data Bank (PDB)

Projects

Single-cell RNA-seq Analysis with Scanpy

Apr 2025 - May 2025

Developed a scanpy pipeline for clustering and differential expression of scRNA-seq data.

Skills: Scanpy, Python, Anndata, Seaborn

ChIP-seq Data Analysis and Motif Discovery

Apr 2025 - May 2025

Performed peak calling using MACS3 and conducted motif discovery to identify potential transcription factor binding sites.

Skills: ChIP-seq, MACS3, R, Python

Bulk RNA-seq Data Analysis

Mar 2025 - April 2025

Built a reproducible RNA-seq analysis pipeline using Nextflow, including quality control, normalization, and differential expression analysis.

Skills: RNA seq, Nextflow, Python, Singularity

Starr-seq Transcriptome Web App

Mar 2025 - April 2025

Developed a web interface for visualizing *Drosophila* transcriptome data from Starr-seq experiments.

Skills: HTML/CSS/JS, Flask, Starr-seq data, SQL

Analysis of Protein Interactions in Triple Negative Breast Cancer

Sep 2022 - Jan 2023

- Molecular docking simulation of Protein-ligand pairs involved in TNBC

Study on Adsorption of Nitrate using Zeolite

Sep 2022 - June 2023

- Experiments for adsorption of Nitrate by Iron-coated Zeolite

Technical Skills

- **Languages & Tools:** Python, R, Linux, SQL, Git, Bash, HTML/CSS/JS
- **Bioinformatics Tools:** Seurat, Scanpy, MACS2, DESeq2, Bioconductor, ggplot2
- **Data Types:** RNA-seq, ChIP-seq, scRNA-seq, Proteomics, Metabolomics
- **Workflows:** Nextflow, GitHub
- **Visualization:** matplotlib, seaborn, ggplot2, volcano plots, PCA, heatmaps

Publications

Sono-assisted adsorption of nitrate by Fe-modified chitosan-zeolite composite material	Mar 2024
<i>Gopal Italiya, Anushka Dongaonkar, Nithyashree Jayakumar, Sangeetha Subramanian</i>	
10.1080/01932691.2024.2325390	
Anti-aging Integrative Therapy Effects: A Clinical Study	Aug 2024
<i>Alok Sharma, Hemangi M. Sane ... , Anushka Dongaonkar</i>	
10.20944/preprints202407.1110.v1	
Evaluating the Impact of Anti-Aging Integrative Therapy on Cellular Aging and Quality of Life	Oct 2024
<i>Alok Sharma, Hemangi M. Sane ... , Anushka Dongaonkar</i>	
10.21275/SR24928103739	